

Aryan Ashraf

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Technical Skills

Languages: TypeScript, C#, Python, Java, SQL, JavaScript, C, C++

Frameworks & Libraries: React, .NET, FastAPI, PyTorch, TensorFlow, YOLOv8, Gemini Vision, Unity

Cloud & Tools: Azure (SQL Database, DevOps), Google Cloud (Cloud Run), CI/CD, Docker, Linux, Git, GitHub, REST APIs, SQL Server, xUnit, Firebase/Firestore (NoSQL)

Concepts & Practices: Unit & Integration Testing, Code Review, Agile/Scrum, Data Structures & Algorithms, Object-Oriented Programming, LLM/Agent Systems, RAG

Experience

Brock Solutions

April 2026 – September 2026

Software Developer Intern | TypeScript, C#, LLM Agents, React, .NET

Kitchener, ON

- Built 10 REST API endpoints, .NET BFF controllers, and a React (TypeScript) client for a ServiceNow case-management portal with OAuth2 and account-scoped authorization for 10+ airline clients.
- Shipped 20+ pull requests across 5 repositories and 2 source control systems, including a config-driven integration suite exercising live ServiceNow REST APIs.
- Designed a KPI trending dashboard for an airport baggage-handling platform covering 10 metrics across 7 time ranges, adding conditional T-SQL time bucketing to 7 .NET metric sources.
- Cut dashboard load latency by deduplicating repeated metric queries and parallelizing under a semaphore capped at 8, validated with 7 dedup and concurrency tests in Azure DevOps CI/CD.
- Architected a multi-agent LLM system with retrieval-augmented generation (RAG), an orchestrator over 5 subagents across 4 internal repositories to automate cross-repo analysis.

University of Guelph

April 2025 – September 2025

ML & Assistive Devices Research Assistant | C, C#, SQL, Firestore

Guelph, Ontario

- Debugged and extended C# logic in Unity for a VR mirror-therapy app, storing 200+ exercise data points in Firestore and charting progress trends in Excel to track patient improvement.
- Maintained Arduino (C) firmware for an arthritis rehabilitation glove, integrating sensors, real-time control logic, and breadboard wiring; validated stable operation across 20+ participants.
- Analyzed 100+ temperature sensor readings for a diabetic sock project in SQL and Excel, building trend visualizations and applying Isolation Forest anomaly detection to flag abnormal patterns.

Projects

Bridge | TypeScript, React, FastAPI, Python, Google Cloud Run, Gemini Vision

- Co-developing a full-stack donation platform as an early-stage startup built at a 24-hour hackathon, pairing a React (TypeScript) and Tailwind front end with 31 FastAPI endpoints on Google Cloud Run, backed by Firestore and Firebase Auth across 3 user roles.
- Built live delivery tracking with Leaflet/OSRM routing and in-app chat, and integrated Gemini Vision to turn a donation photo into a structured item list, cutting listing time from about 2 minutes to under 10 seconds.

Real-Time ASL Detection Program | Python, PyTorch, TensorFlow, YOLOv8, Labellmg, Jupyter

- Built a real-time American Sign Language detection system in Python with PyTorch, TensorFlow, and YOLOv8, classifying 10 signs at 85%+ accuracy through a live video interface.
- Annotated and managed a 1,500-image dataset with Labellmg, creating 80/20 train/test splits for model training and validation.

Education

University of Guelph

Expected April 2028

Bachelor of Engineering in Computer Engineering (Honours)

Guelph, Ontario

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Autonomous Systems & Robotics

Extracurricular

UofG Robotics Team: Cut rover joystick-to-motor latency by replacing ROS2 layers with C++ UDP and CAN bus.

Focus On Engineering Guelph: Built and maintained a textbook rental website and ran engineering competitions.